GNSS Spoofing Detection and Recovery

Implementation Guide v2 - modular code edition

Based on Kujur, Khanafseh, Pervan (2024), NAVIGATION 71(1). Reference code: the modular MATLAB files (protected_nav_step.m and below). Quick checklist: QUICKSTART_integration.m.

1. What the system does

DETECT the attack, LATCH the alarm, navigate on a clean INS-only solution (COAST), detect the END of the attack, test GNSS on a trial filter (PROBATION), HANDBACK when proven clean.

From the paper	Added by us (the paper leaves these open)
CPI monitor: detects the spoofer tracking error. Chi-square on position-domain innovations (Eq. 17-38).	Alarm LATCH: monitor silence never ends the alarm. Needed because the monitors go quiet about 20 s into a ramp (capture) while the attack continues.
SS monitor: filter vs a short INS-only coast, one per window (Eq. 49-52). Gives real-time protection levels.	End-of-attack test (q_reval): chi-square of raw GNSS against the clean coast. Huge while spoofed, drops the epoch the attack stops.
Overlapping windows (Fig. 9): one opens every epoch, lives N epochs.	PROBATION: a trial filter takes GNSS under monitor watch while the output stays on the coast. Any alarm vetoes the handback.
Recovery rule: fall back to the coast of the last alarm-free window (Section 5).	3-axis bank: one monitor per position axis, latch on any axis.

2. The files

File	Job
STRUCT CLASSES: KujurConfig.m, NavState.m, PoolState.m, NavMode.m, NavInfo.m, PoolReport.m, SsResult.m	Definitions only, no logic. Every struct built in one place: cfg = KujurConfig.create(...), sys = NavState.create(...), pool = PoolState.create(...); mode constants in NavMode; blank telemetry / report / result factories in the rest.
protected_nav_step.m	The brain: the 3-mode state machine. One call per GNSS epoch.
monitor_pool_step.m	Runs all open windows: SS every epoch, CPI at close. Reports alarms and the anchor candidate.
monitor_pool_open_window.m, monitor_pool_clear.m	Open / reset the windows.
ss_monitor_step.m	SS test for one window, all axes (Eq. 49-52).
cpi_monitor_window.m	CPI test at window close (Eq. 33/35). Original file, unchanged.
revalidation_test.m	q_reval: is GNSS consistent with the clean coast?
ins_coast_step.m	INS-only propagation of a state (E28/E31).
kalman_update_step.m	One KF epoch. In your system: REPLACE with your own EKF.
spoof_monitor_2hz.m	YOUR 2 Hz wrapper template: persistent sys/cfg/epoch, reset input, mode flag and reset command outputs for the 100 Hz side.
ins_100hz_template.m	YOUR 100 Hz additions: Phi_acc/Q_acc accumulators, feedback gate, reset consumption.
recovery_nav_sim.m	Thin demo loop over logged data (harness).
solve_N_min.m, compute_PMD_eq38.m, kujur_params.m, build_Phi_Q.m	Offline design tools. Desktop only.
ss_monitor_update.m, dual_monitor.m, test_all.m	Legacy single-axis versions. Kept because test_all validates the math.

Call tree:

```
recovery_nav_sim (or spoof_monitor_2hz) -> protected_nav_step -> kalman_update_step,
monitor_pool_step (-> ss_monitor_step, cpi_monitor_window),
monitor_pool_open_window, ins_coast_step, revalidation_test
```

3. Before you start - three checks

1. Find the line where your EKF feeds corrections back into the INS (you will add an ON/OFF flag there).
2. Find your position state indices in NED (example: N=1, E=2, D=3).
3. Sign test: add +10 m to a test measurement along H(:,idxD). The estimate at idxD must move +10 m. This catches wrong index, frame mix-up, and sign errors in one shot.

4. What your EKF must provide (every GNSS update)

```
gamma = z - h(x_bar) the real innovation
S = H*P_bar*H' + V
H, x_hat, P_hat, Phi, Q, z, V
```

Rule: gamma must be the real innovation from your filter. Synthetic residuals break the SS math (Eq. 50).

5. Offline design - once, then hard-code

Replay a clean run that includes your WORST sky (masking, low satellite count). Then, per axis a = 1, 2, 3:

```
s2(k) = f' * (S \ f), f = H(:,a) every epoch
s2min(a) = min over all k worst geometry
N_a(a) = solve_N_min(s2min(a), 0.10, PFAp/3, PFAm/3, 1e-7, 1e-6)
```

Then the shared constants:

```
N = max(N_a) + 1 (+1 = margin)
T_N = gaminv(1 - PFAp/3, N/2, 2)
k_FA = norminv(1 - (PFAm/3/N)/2) two-sided
k_MD = norminv(1 - 1e-6) one-sided
T_reval(m) = gaminv(1 - 1e-3, m/2, 2) per satellite count
```

Hard-code everything. Never compute on the target: N sizes arrays at compile time, and the design needs the whole mission envelope, not the sky at boot. Re-run only when the IMU, constellation set, or budgets change. Runtime extra (one line): if s2(k) < s2min, raise a "coverage degraded" advisory.

6. Runtime - the three modes

Mode	Output	What runs	Exit
NOMINAL	your EKF	EKF update. monitor_pool_step. Refresh the anchor when a window closes clean on an alarm-free epoch. Open one new window.	Any alarm: LATCH. Reset nav to the anchor (bring it to now with ins_coast_step in a loop), clear pool, go to COAST.
COAST	INS only	ins_coast_step. revalidation_test on the raw GNSS. Count consecutive passes (dwell).	10 passes in a row: go to PROBATION (trial = copy of the coast). Monitor silence NEVER exits this mode.
PROBATIO N	still INS only	Trial filter takes the GNSS update. monitor_pool_step watches THE TRIAL. Output keeps coasting.	Alarm: VETO, back to COAST. N+8 quiet epochs: COMMIT - output = trial, anchor = trial, back to NOMINAL.

The 100 Hz INS function needs only two additions: a feedback ON/OFF flag (OFF in COAST and PROBATION) and a state-reset command (used at latch and at commit). All statistics and decisions live in the 2 Hz function.

7. Parameters (values used in the demo)

Name	Value	Meaning
N (window_length)	max per-axis N_min + 1	Epochs per window. Sets CPI missed-detection performance.
T_N (cpi_threshold)	gaminv at N	CPI alarm gate.
k_FA (k_false_alert)	about 5.2	SS gate multiplier (two-sided).
k_MD (k_missed_detection)	about 4.75	Protection-level multiplier (one-sided).
T_reval	about 39 (chi-square, m=16)	GNSS-vs-coast pass gate. Table per satellite count.
M_dwell	10 epochs (5 s)	Consecutive passes to open probation.
M_prob	N + 8 epochs	Quiet epochs to commit. Floor is N (one full CPI window). The +8 is margin: several CPI votes, and cover for a spoofer that resumes late.
Max coast time	your IMU drift budget	After this, declare dead-reckoning. Not optional.

8. Rules that must never break

1. The alarm is sticky. Monitor silence never exits COAST - the monitors go quiet during capture while the attack continues.
2. During PROBATION the output never ingests a GNSS measurement. Only the disposable trial does.
3. The anchor is refreshed only when a window closes clean on an epoch with no alarm anywhere.
4. Enforce the max coast time. The coast is a perishable reference; IMU grade sets its shelf life.

9. Test order

1. Shadow mode (monitor only, no authority) on recorded clean data: expect zero alarms, and s2(k) never below s2min.
2. Scripted attacks - dither only, ramp only, both, oblique direction: expect latch within 1 s, error always inside 3-sigma, every premature probation vetoed, exactly one commit.
3. Slow-ramp sweep: find the detection floor per axis and document it. Attacks below the floor never latch - that is the honest limit.
4. Give the state machine output authority. Repeat step 2.
5. Coder refactor last (fixed sizes, precomputed thresholds).

10. Known limits

1. Drags slower than the detection floor never trip an alarm, so recovery never engages.
2. Coast time is limited by INS drift. Navigation grade: minutes. Automotive: tens of seconds.
3. Innovation-based monitors cannot see a captured filter. Only the clean-coast reference (q_reval) and independent sensors (for example baro on the vertical) can. This is why the latch exists.